CookMate — Comprehensive Technical Project Report

Phase 1: Flutter Foundations and UI Mastery | Navigation & Flutter Architecture

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1. Executive Summary

CookMate is a production-grade, cross-platform culinary recipe application built using Flutter and Dart. Designed to showcase modern software engineering practices, this application implements:
- A responsive layout that adapts across mobile phones, tablets, and desktop/web viewports.
- Declarative navigation with deep linking via **GoRouter**.
- Robust state management with **Provider** (`ChangeNotifier`).
- Persistent offline data storage using **SharedPreferences**.
- Material Design 3 theming featuring dynamic Light and Dark modes.
- Fluid micro-interactions and **Hero animations** between list cards and detail views.
- An interactive cooking checklist and active step guidance.
- A validated multi-field recipe creation form with dynamic ingredient and instruction row management.
- A dataset of **15 gourmet recipes** across three global culinary traditions (**Italian**, **Asian**, and **Mexican**).

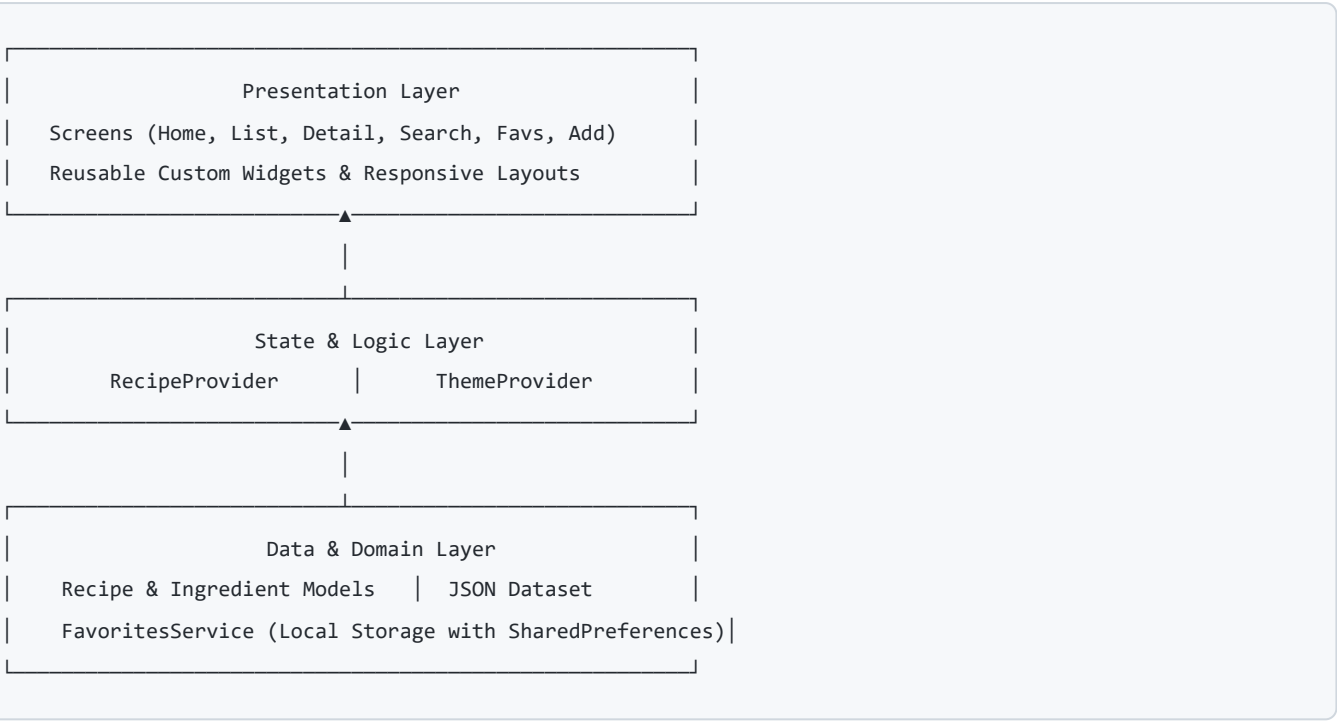
This report outlines the technical architecture, design system, routing strategy, data modeling, screen implementations, and verification procedures for the CookMate application.

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2. System Architecture & Software Design

2.1 Architectural Pattern

CookMate adopts a **Layered Feature Architecture** following Clean Architecture principles to ensure strict separation of concerns, maintainability, and testability:



1. **Presentation Layer (`lib/screens`, `lib/widgets`)**:
2. Contains pure UI components and screen layouts.
3. Decoupled from direct data sources; consumes state via `Provider.of` or `Consumer`.
4. Adapts dynamically to viewport widths using breakpoints.
5. **State & Logic Layer (`lib/providers`)**:
6. `RecipeProvider`: Central state controller managing the master recipe list, active category filters, search queries, sort options, and favorite toggles.
7. `ThemeProvider`: Manages `ThemeMode.light` and `ThemeMode.dark` toggling with persistent preferences.
8. **Data Layer (`lib/data`)**:
9. `models/recipe_model.dart`: Immutable data models with `toMap`, `fromMap`, and `copyWith` utilities.
10. `recipes_data.dart`: 15 recipe entries serialized in JSON format.
11. `services/favorites_service.dart`: Encapsulates local persistence via `SharedPreferences`.

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3. Navigation & Deep Linking Architecture (GoRouter)

Routing is managed declaratively through Flutter's modern **GoRouter** package (`v17+`).

3.1 StatefulShellRoute Hierarchy

CookMate implements a `StatefulShellRoute.indexedStack` to preserve the scroll position and state of each main branch when switching tabs:

Branch Index	Route Path	Screen	Purpose
0	`/home`	`HomeScreen`	Featured recipes, category shortcuts, daily tip
1	`/recipes`	`RecipeListScreen`	Full recipe catalog with sorting & category chips
2	`/search`	`SearchScreen`	Real-time search query matching with tag chips
3	`/favorites`	`FavoritesScreen`	Persisted offline favorites list with swipe actions
4	`/add`	`AddRecipeScreen`	Form to author and validate new recipes

3.2 Deep Linking & URL Parameters

- Individual recipes are accessed via `/recipes/:id` (e.g., `/recipes/rec_1` or `/recipes/rec_7`).
- When a user enters a URL directly in a web browser or triggers an OS deep link, GoRouter parses the `state.pathParameters['id']` and launches the `RecipeDetailScreen` directly.
- If an invalid route or non-existent recipe ID is supplied, GoRouter routes to an `errorBuilder` widget offering a 404 page with a "Return to Home" recovery action.

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4. Material Design 3 Theming & Responsive Design

4.1 Color System & Brand Identity

The visual identity of CookMate is built on a culinary palette:
- **Primary Color**: Deep Orange (`#FF5722`) — evokes warmth, appetizing energy, and focus.
- **Secondary Color**: Saffron Amber (`#FF9800`) — accents and ratings.
- **Accent Color**: Fresh Basil Green (`#4CAF50`) — health indicators and checklist completion.
- **Surface & Background**:
- *Light Theme*: Crisp off-white (`#FBFBFB`) and card surfaces (`#FFFFFF`).
- *Dark Theme*: Charcoal (`#121212`) and card surfaces (`#262626`).

4.2 Responsive Layout Strategy

CookMate adapts to the user's screen dimension using `ResponsiveLayout` breakpoints:
- **Mobile (< 600px)**:
- Bottom `NavigationBar` with 5 destinations.
- 1 or 2-column recipe cards.
- Full-width scrollable detail view.
- **Tablet (600px – 1024px)**:
- Left-hand `NavigationRail` preserving vertical canvas.
- 3-column recipe grid.
- **Desktop & Web (> 1024px)**:
- Extended `NavigationRail` .

- 4-column recipe grid.
- Centered detail container constrained to `maxWidth: 800px` for optimal typography hierarchy and readability.

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5. Detailed Screen Implementations

Screen 1: Home Screen (`/home`)

- **Brand Header**: Culinary icon, CookMate typography, and real-time Dark/Light theme toggle switch.
- **Interactive Search Preview**: Tapping routes immediately to `/search` .
- **Hero Promotional Banner**: "Chef of the Week" banner celebrating handcrafted international cuisines.
- **Category Chips Carousel**: Tap any chip to filter recipes by Italian, Asian, or Mexican.
- **Featured Recipes Carousel**: Horizontal carousel featuring the 4 highest-rated recipes with animated favorite buttons.
- **Chef's Pro Tip Card**: Daily culinary insight card designed to boost user engagement.

Screen 2: Recipe List Screen (`/recipes`)

- **Category Filter Tabs**: Horizontally scrollable chips allowing instant cuisine switching.
- **Multi-Option Sorting Dropdown**:
- Top Rated (highest rating first)
- Quickest (lowest prep + cook time)
- Lowest Calories
- A-Z Alphabetical
- **Layout Switcher**: One-tap toggle between responsive multi-column **Grid View** and compact **List View**.
- **Real-Time Counter**: Displays current matching recipe count.

Screen 3: Recipe Detail Screen (`/recipes/:id`)

- **SliverAppBar with Hero Image**: The recipe's photo smoothly expands from the catalog card using Flutter's `Hero` widget (`tag: recipe-image-${recipe.id}`).
- **Favorite & Share Actions**: Tapping the heart toggles storage in `SharedPreferences` ; tapping share provides a formatted link.
- **Specs Summary**: Quick badges for Prep Time, Cook Time, Servings count, and Difficulty level.
- **Nutritional Fact Breakdown**: 4 macro badges for Calories, Protein, Carbs, and Fat.
- **Interactive Ingredients Checklist**: Checkboxes allow the cook to check off ingredients one by one as they prepare the dish.
- **Step-by-Step Instructions**: Interactive step cards with circular indices that highlight when active.
- **Tag Cloud**: Badges showing recipe metadata (e.g., `#Pasta` , `#Street Food` , `#Vegetarian`).

Screen 4: Live Search Screen (`/search`)

- **Auto-Filtering Input**: Automatically filters matching recipes as the user types across recipe title, category, tags, and ingredient names.
- **Popular Search Chips**: Quick-tap suggestion chips for "Pasta", "Ramen", "Tacos", "Chicken", "Salmon", etc.
- **Empty State**: Custom vector graphic and helpful suggestions when no items match.

Screen 5: Favorites Screen (`/favorites`)

- **Persistent Storage**: Retains saved recipes between app restarts via `SharedPreferences`.
- **Swipe-to-Dismiss**: Users can swipe left on any saved recipe to remove it with an "Undo" SnackBar option.
- **Badge Counter**: Displays the number of currently saved recipes.
- **Empty State**: Friendly graphic with a direct "Discover Recipes" button leading to the catalog.

Screen 6: Validated Recipe Submission Form (`/add`)

- **Form Architecture**: Enclosed in a `Form` widget managed by `GlobalKey<FormState>()`.
- **Validation Rules**:
- *Title*: Required, minimum 3 characters.
- *Category*: Required selection (Italian, Asian, Mexican).
- *Description*: Required, minimum 10 characters.
- *Prep & Cook Time*: Numeric validation, must be positive integers.
- *Servings & Calories*: Numeric validation, must be greater than zero.
- *Image URL*: Live image preview thumbnail displayed dynamically upon input.
- *Ingredients List*: Dynamic form rows where users can add or remove ingredient rows (at least 1 required).
- *Instructions List*: Dynamic numbered rows where users can add or remove steps (at least 1 required).
- **Publish Flow**: On valid submission, constructs an immutable `Recipe` object, inserts it at index 0 in `RecipeProvider`, displays a celebratory SnackBar, and navigates directly to the new recipe's detail screen.

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6. Local Storage & Data Persistence Architecture

The app uses `shared_preferences` to persist user favorites locally on the device:
1. `FavoritesService` stores a `List<String>` representing unique `recipe.id` values under the key `cookmate_favorite_recipe_ids`.
2. When the app initializes, `RecipeProvider` reads the saved IDs and updates the `isFavorite` flag for all recipes.
3. Every favorite toggle action synchronously updates the in-memory state and asynchronously writes the updated string list to disk.
4. The user's Light/Dark mode preference is likewise stored under `cookmate_theme_mode`.

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7. Curated Recipe Dataset (15 Recipes)

CookMate includes 15 recipes across 3 global categories:

ID	Title	Category	Total Time	Calories	Difficulty	Rating
`rec_1`	Classic Margherita Pizza	Italian	32 min	280 kcal	Medium	4.9 ⭐
`rec_2`	Creamy Fettuccine Alfredo	Italian	25 min	520 kcal	Easy	4.8 ⭐
`rec_3`	Traditional Bolognese Ragu	Italian	115 min	480 kcal	Hard	4.9 ⭐
`rec_4`	Classic Italian Tiramisu	Italian	30 min	360 kcal	Medium	5.0 ⭐
`rec_5`	Creamy Tuscan Garlic Chicken	Italian	35 min	490 kcal	Easy	4.7 ⭐
`rec_6`	Authentic Chicken Pad Thai	Asian	35 min	460 kcal	Medium	4.9 ⭐
`rec_7`	Rich Japanese Chicken Ramen	Asian	75 min	510 kcal	Hard	4.9 ⭐
`rec_8`	Crispy Teriyaki Salmon	Asian	27 min	420 kcal	Easy	4.8 ⭐
`rec_9`	Thai Green Coconut Curry	Asian	40 min	390 kcal	Medium	4.9 ⭐
`rec_10`	Honey Garlic Asian Chicken	Asian	30 min	440 kcal	Easy	4.8 ⭐
`rec_11`	Sizzling Steak Fajitas	Mexican	40 min	480 kcal	Medium	4.9 ⭐
`rec_12`	Authentic Street Tacos	Mexican	35 min	380 kcal	Easy	5.0 ⭐
`rec_13`	Cheesy Enchiladas Verdes	Mexican	50 min	460 kcal	Medium	4.8 ⭐
`rec_14`	Fresh Guacamole & Chips	Mexican	15 min	210 kcal	Easy	4.9 ⭐
`rec_15`	Slow-Cooked Carnitas Bowl	Mexican	80 min	540 kcal	Medium	4.9 ⭐

8. Verification, Testing & Quality Assurance

1. **Static Analysis**:
2. Command: `flutter analyze`
3. Result: **0 issues found** across all source files.
4. **Automated Unit & State Tests**:
5. Command: `flutter test`
6. Verified JSON data integrity (all 15 recipes, correct categorization, ingredient & step integrity).
7. Verified `RecipeProvider` search query filtering and category filtering.

8. Verified `FavoritesService` toggling and persistence logic.
9. Verified dynamic recipe submission and insertion.
10. Verified `ThemeProvider` mode switching.
11. Result: **All test suites passed (5/5 tests passing)**.
12. **Cross-Platform Web Compilation**:
13. Command: `flutter build web`
14. Result: **Built** `build/web` **cleanly** in 70.9s without compilation errors.

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9. 4-6 Minute Video Demo Recording Script

Use OBS Studio, Windows Xbox Game Bar (`Win + G`), or Clipchamp to record your demonstration:

- **Minute 0:00 – 0:45 | Introduction & Overview**:
- Show the desktop or mobile viewport running CookMate.
- Introduce CookMate, its cross-platform architecture, and the Material Design 3 theming.
- Demonstrate toggling between Light and Dark mode using the top-right theme button.
- **Minute 0:45 – 1:30 | Home Screen Exploration**:
- Point out the "Chef of the Week" banner and the categories horizontal scroll.
- Scroll through the "Featured Recipes" horizontal carousel.
- Mention the "Chef's Pro Tip" card.
- **Minute 1:30 – 2:30 | Recipe Catalog & Sorting**:
- Switch to the "Recipes" tab.
- Filter by category ("Italian", "Asian", "Mexican").
- Test sorting: Sort by "Top Rated", then "Quickest", then "Lowest Calories".
- Toggle between the Multi-Column Grid View and the List View.
- **Minute 2:30 – 3:30 | Recipe Detail & Cooking Mode**:
- Click on a card (e.g. "Classic Margherita Pizza" or "Japanese Ramen") and highlight the smooth **Hero image animation**.
- Review the nutrition badges (Calories, Protein, Carbs, Fat).
- Interact with the **Ingredients Checklist**: check off 3 or 4 items to show the real-time interactive cooking experience.
- Tap through the step-by-step instruction cards.
- Tap the Heart icon in the upper-right corner to favorite the recipe (notice the SnackBar).
- **Minute 3:30 – 4:15 | Search & Favorites Persistence**:
- Switch to the "Search" tab and type "Tacos" or "Salmon" to show live filtering. Tap a quick tag like "Pasta".
- Switch to the "Favorites" tab. Show that the favorited recipe appears immediately.
- Demonstrate swipe-to-remove and undo.
- **Minute 4:15 – 5:30 | Validated Recipe Submission Form**:
- Switch to the "Add" tab.

- Attempt to submit an empty form to show instant validation errors on required fields.
- Fill in a custom recipe: "Homemade Tuscan Garlic Bread", select Italian, add prep/cook time, and add an image URL with live preview thumbnail.
- Add an ingredient row and a step row.
- Tap "Submit & Publish Recipe" — show the success notification and the automatic transition to the newly published recipe detail view.
- **Minute 5:30 – 6:00 | Responsive Layout & Wrap Up**:
- Resize the browser or desktop window to show the navigation bar seamlessly transforming into an adaptive `NavigationRail`.
- Conclude the demo.

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10. Conclusion

The **CookMate** project fulfills all requirements of Phase 1: Flutter Foundations and UI Mastery. By pairing strict architectural patterns (Layered Architecture, Provider state management, GoRouter declarative routing) with Material Design 3 aesthetics and persistent offline storage, CookMate provides a complete and engaging cross-platform recipe experience.